Local causal trials without global equilibration

A finite thermal preparation and measurement construction

Research construction and independent companion results

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Completed result and its meaning

The strongest completed construction is a **local causal-trial operation on a retained nonlinear thermal apparatus**. A fresh local Langevin thermostat, finite excess relative entropy and a private physical clock can supply a known random momentum ensemble approximately independent of the contemporaneous hidden environment and old classical reference. A finite pointer instrument implements the required measurement with explicitly related imprecision, backaction, initialization, timing, range and control costs.

The operation does not identify the hidden dynamics, acquire a response table, restore the hidden equilibrium ensemble, or supply an arbitrary momentum overwrite. It makes a specified local randomized experiment scientifically interpretable despite unknown hidden dynamics. The hidden distribution shared by the trial arms is whatever the passive waiting and measurement window produces. An ordinary bounded weighted score is preferable when the subsequent experiment does not use an arm label.

The completed theorem is conditional science within the stated classical Hamiltonian/Langevin and instrument contracts. Its mathematical mechanism uses established conditional entropy and Gaussian logarithmic Sobolev machinery. Historical priority, a laboratory demonstration and civilisation-scale impact are not established. The consequence earned here is the complete reference-sensitive physical instrument and its separation from global mixing.

1. Physical and information resources

Write the full physical state as $S = (p, z)$, where z contains the coordinate x conjugate to p and every hidden degree of freedom. Assume a well-posed Hamiltonian baseline with

$$H(p,z) = \frac{p^2}{2M} + V(z), \qquad \pi(dp, dz) = G_\sigma(dp)\pi_z(dz), \qquad \sigma^2 = M/\beta.$$

Here G_σ is the centered Gaussian of variance σ^2 and π is a normalizable invariant probability law. The conservative dynamics preserves phase volume and H. An actual local thermostat acts as

$$dx = \frac{p}{M}dt, \qquad dp = F(S)dt - \gamma p dt + \sqrt{2\gamma\sigma^2}\, dW_t, \quad \gamma > 0.$$

Additional invariant dissipative channels are allowed if they cannot increase relative entropy. Hidden nonlinear interactions and an unknown finite hidden dimension are allowed. An infinite-dimensional extension requires the same well-defined invariant measure and entropy identities; it is not granted merely by writing infinitely many coordinates. Momentum-dependent interactions that destroy the displayed factorization are outside this statement.

Let R be all pre-existing retained classical information. Supply the bound

$$D(\rho_{SR} \| \pi \otimes \rho_R) \le D_0 < \infty. \tag{1}$$

This includes both physical nonequilibrium free energy and shared information:

$$D(\rho_{SR} \| \pi\rho_R) = D(\rho_S \| \pi) + I(S:R).$$

It is not a total-energy bound. Continuous equilibrium thermal input is allowed. D_0 can be extensive, so absence of a separate dimension factor is not a claim that a highly excited large apparatus is cheap. A quiet local trace does not certify (1). A preparation protocol, supplied ensemble contract or other valid thermodynamic bound must supply it. An exact reference copy of continuous microscopic data can make it infinite. Future thermostat noise is fresh and unobserved, independent of the initial source and reference.

The deployment observer has local measurement and receiver controls, a private timer and initialized receiver memory. A momentum-reset actuator, hidden-state access and a response model are not supplied. Known temperature, friction, mass, meter gains, force/moment bounds and a finite validity range are substantive calibration resources. The operation does not infer them without exposure.

2. The local replacement theorem

Define the mathematical replacement C to retain the joint (z, R) marginal and draw p independently from G_σ. Put

$$J(t) = D(\rho_{SR,t} \| C\rho_{SR,t}).$$

Theorem. Under the preceding contracts,

$$\int_0^\infty J(t)\,dt \le \frac{D_0}{2\gamma}. \tag{2}$$

If T is an independent private trigger with density $g(t) \le C_g$, then, with $\mathrm{TV} = \frac{1}{2}\|P - Q\|_1$,

$$\mathrm{TV}(\mathcal{L}(T, S_T, R),\ G_\sigma \otimes \mathcal{L}(T, z_T, R)) \le \sqrt{\frac{C_g D_0}{4\gamma}}. \tag{3}$$

This includes the actual time and the full hidden/reference state. It is an average guarantee over private timing and old history, not a guarantee conditioned on every exceptional clock value or posterior.

Proof. Let $h_t = d\rho_{SR,t}/d(\pi\rho_R)$ and $I_p(t) = \mathbb{E}|\partial_p \log h_t|^2$. Integration by parts gives $-D'(t) \ge \gamma\sigma^2 I_p(t)$: the conservative generator contributes zero and the thermostat contributes its Dirichlet form. Conditional on each (z, R), Gaussian logarithmic Sobolev gives $J(t) \le \sigma^2 I_p(t)/2$. Integration proves (2). Applying the entropy chain rule to the time-labeled laws gives relative entropy at most $C_g D_0/(2\gamma)$; Pinsker's inequality proves (3). Finite-entropy initial laws use the integrated entropy identity and its standard approximation form. No spatial logarithmic Sobolev constant or global mixing estimate is used.

A useful deduction is the exact decomposition

$$J(t) = D(\rho_{p,t}\|G_\sigma) + I(p_t : z_t, R). \tag{4}$$

Local dissipation therefore budgets both local nonthermal shape and hidden confounding. A system can fail to approach its full equilibrium while still supplying the local resource in (3).

3. Finite private timing

Use an isolated Poisson arrival clock of calibrated rate r. Accept the first arrival before a finite deadline H; otherwise report timeout. Conditional on arrival, its density is bounded by

$$C_g = \frac{r}{1 - e^{-rH}}, \qquad \Pr(\text{timeout}) = e^{-rH} = \alpha_c.$$

For thermal error allocation δ_{th}, sufficient choices are

$$r = \frac{4\gamma(1-\alpha_c)\delta_{\text{th}}^2}{D_0}, \qquad H = \frac{D_0 \log(1/\alpha_c)}{4\gamma(1-\alpha_c)\delta_{\text{th}}^2}. \tag{5}$$

Expected clock occupancy including timeout is $(1-\alpha_c)/r$. Fixed finite meter duration and independent latency shift or convolve the clock density without increasing its cap. Fine event time need not be digitized. A bounded clock-law discrepancy adds its own TV allowance. The $D_0 = 0$ case is already locally ready and does not require division by zero in (5).

A finite-intensity coherent optical pulse with photon counting is one candidate clock architecture under its actual Poisson/detector contract. Detection efficiency η_d requires photon flux r/η_d and nominal optical energy $\hbar\omega r H/\eta_d$. Laser, detector, watchdog and controller standby power over H are additional. Dark counts, afterpulsing, latency, thermal coupling and any information leak into

the specimen require calibration. This is an explicit timing resource, not a claim that every photon detector automatically meets it.

An arbitrary finite point grid does not follow from (2), because an integrable quantity may concentrate at the grid times. A digital clock needs an additional time-regularity proof or actual independent continuous dither. At timeout the protocol aborts; it does not launch at that atom and reuse (3).

4. From local readiness to a causal trial

Choose a known odd Lipschitz score $s : \mathbb{R} \to [-1, 1]$, and assign

$$\Pr(A = a \mid p) = \frac{1 + as(p)}{2}, \qquad q_a(dp) = G_\sigma(dp)[1 + as(p)], \quad a \in \{-1, +1\}.$$

Under exact replacement, A is fair and independent of (z, R, T), while $p \mid A = a$ has law q_a independently of those variables. Thus the marked joint law is exactly the law of first choosing a fair arm and then preparing its known momentum distribution while preserving the passive hidden marginal. All runs are used. Marking and every common subsequent experiment are stochastic channels, so (3) transfers to their complete joint law without a future-horizon multiplier.

The sinusoidal choice $s(p) = \sin(p/\sigma)$ gives

$$\mathbb{E}_{q_\pm} p = \pm\sigma e^{-1/2}, \qquad \mathbb{E}_{q_\pm} p^2 = \sigma^2.$$

For the same Lipschitz constant, clipped-linear $s(p) = \mathrm{clip}(p/\sigma, -1, 1)$ gives the larger mean $\pm\sigma[2\Phi(1) - 1] \approx \pm 0.682689\sigma$, with the same kinetic second moment. It is implementable by finite arithmetic and comparisons. This is a useful ordinary improvement, not the central consequence.

The output is a randomized arm, not deterministic control of momentum. For an externally requested arm, accept with the corresponding probability; ideal success is $1/2$, and conditioning increases an unconditional TV error ϵ to at most 2ϵ. Failures consume their actual time and resources. For general rare accepted targets the acceptance probability must be paid. Since $q_a/G_\sigma \le 2$, labeling itself cannot accelerate the production of a rare outcome.

Fine current preparation readings are excluded from the prepared output. They are correlated with p and cannot simultaneously be treated as an independent old reference. Keep the arm, old reference, declared clock record and future reports. Temporary readings must be genuinely erased or isolated. They cannot control later timing through a hidden data-dependent latency.

5. The finite meter and its physical backaction

Take two identical weighted position windows of duration b, separated by ℓ, and set $t_* = \ell + b$. The estimator is $\widehat{p} = M(Y_1 - Y_0)/\ell$. The exact noiseless expression is a positive normalized average of momentum over the window. Use the momentum at its END as the logical intervention variable. The trigger for that endpoint remains bounded-density by the fixed time shift.

Supply local moments $\mathbb{E}p^2 \le P_*^2$ and $\mathbb{E}|F - \gamma p|^2 \le B_*^2$. For an unbounded coordinate, supply $\mathbb{E}x^2 \le X_*^2$ or an appropriate range contract. For example, Gaussian kinetic structure and (1) imply

$$P_*^2 = 4\sigma^2[D_0 + (\log 2)/2].$$

A bounded local conservative force gives $B_* \le F_* + \gamma P_*$. A local equilibrium exponential force moment gives another entropy-variational bound. These are metrology premises; factorization alone does not bound an arbitrary force or an unbounded position.

Finite sensor error e_x per position window and white detector density ν_x give the sufficient mean error

$$e_p \le B_* t_* + \sqrt{2\gamma\sigma^2 t_*} + \frac{2Me_x}{\ell} + M\sqrt{\frac{2\nu_x}{b\ell^2}}. \tag{6}$$

Sensor clearing, integration, ADC, timing and response lag enter these terms or a separate overflow allowance. A mark with Lipschitz constant L_a changes the joint law by at most $L_a e_p$. No independence of meter error and later trajectory is assumed. A native momentum-box readout can avoid differentiation where the actual port supplies it, with its own loading/noise contract.

5.1 Explicit bounded pointer interaction

For each window initialize a fresh canonical pointer (Q, P) with mass m_d and centered variances v_Q, v_P. Use the smooth Hann weight

$$w(t) = \frac{1 - \cos(2\pi t/b)}{b}, \qquad 0 \le t \le b,$$

and the bounded interaction

$$H_{\text{int}}(t) = -Gw(t)\chi_x(x)\chi_P(P). \tag{7}$$

The smooth saturations equal the identity on declared finite ranges and obey $|\chi'| \le 1$, $|\chi_x| \le X_{\text{cap}}$ and $|\chi_P| \le P_{\text{cap}}$. The pointer is free during the window. On the good range event,

$$Q(b) = Q_0 + \frac{bP_0}{m_d} - G\int_0^b w(t)x(t)dt,$$

so measuring the detached pointer gives a position average with error

$$e_x \le \frac{\sqrt{v_Q}}{G} + \frac{b\sqrt{v_P}}{m_d G} + \frac{e_{\text{read}}}{G}. \tag{8}$$

Pointer momentum is constant. Its source disturbance is only in the noisy momentum direction: $u = Gw\chi_x'\chi_P$. For two pointers,

$$\mathbb{E}\int u^2 dt \le \frac{3G^2 v_P}{b} =: J_u, \quad \delta_{\text{back}} \le \sqrt{\frac{J_u}{8\gamma\sigma^2}}. \tag{9}$$

The latter is a full-path Girsanov/Pinsker comparison, including hidden state and old reference, under the stated absolute-continuity conditions. Saturated pointer momentum also gives a pathwise action bound. A small displacement perturbation alone would not supply this arbitrary-hidden-state TV guarantee.

All controls in (7) are finite:

$$|H_{\text{int}}|_{\max} \le \frac{2GX_{\text{cap}}P_{\text{cap}}}{b}, \quad |u|_{\max} \le \frac{2GP_{\text{cap}}}{b}, \quad |\dot{g}|_{\max} \le \frac{2\pi G}{b^2}.$$

Hann switching work per box is at most $4GX_{\text{cap}}P_{\text{cap}}/b$, in addition to driver and reset costs. On the passive comparison path, source-range failure is at most $[2X_*^2 + 2t_*^2P_*^2/M^2]/X_0^2$. Gaussian pointer tails and a finite Q-ADC range add their explicit allowances. Compare the saturated-meter path to the passive path first, then evaluate the same readout functional; this avoids charging a second backaction comparison just for overflow. A rotor requires a separately earned angular chart/encoder; (7) is a real-coordinate meter.

5.2 Pointer initialization and physical limits

While isolated from the source, a known harmonic pointer with $k_d = \theta_d/v_Q$, $m_d = v_P/\theta_d$ and a calibrated local bath of thermal energy θ_d has the target Gaussian covariance $\Sigma = \operatorname{diag}(v_Q, v_P)$. For its known drift matrix A_d, put $\rho(t) = \|\Sigma^{-1/2}e^{A_d t}\Sigma^{1/2}\|$. If the initial pointer, including correlations with any old reference, has normalized second moment E_d, its reset joint TV is at most $\rho(t)\sqrt{E_d+1}/2$ when $\rho \le 1/2$. At critical damping, $\rho(t) \le 2e^{-\omega_d t/2}$, giving the finite reset time

$$t_{\text{reset}} \ge \frac{2}{\omega_d}\log\frac{2}{\rho_*}, \quad \rho_* = \min\{1/2, 2\delta_d/\sqrt{E_d+1}\}. \tag{10}$$

This follows from the exact Gaussian transition covariance and entropy convexity, not a free fresh-pointer assumption. Use actual available damping if critical damping is unavailable. Trap release follows a finite known schedule; propagate the covariance through it before using (8)–(9). Refrigeration, trap preparation, initial energy, reset heat, readout and ramp work are actual additional costs. The proof supplies bath-contact time, not a refrigerator operating for free at a thermodynamic lower bound.

Imprecision and backaction are linked by (8)–(9). A quantum pointer must satisfy $v_Q v_P \ge \hbar^2/4$, and a classical realization needs a comfortably larger phase-space scale. Fixed hardware therefore has an error floor. The classical theorem cannot be continued to arbitrarily small error through quantum or finite-bandwidth limits. The accessible bounded canonical coupling in (7) is an engineered meter resource; a named off-the-shelf probe is not assumed to provide it automatically.

6. Complete error and resource law

For a finite L-bit Bernoulli coin and a finite numerical mark, conditional on the declared clock-success branch,

$$\epsilon_{\text{one}} \le \sqrt{\frac{C_g D_0}{4\gamma}} + L_a e_p + \delta_{\text{back}} + \eta_{\text{overflow}} + \eta_{\text{clock}} + \eta_{\text{num}} + 2^{-L} + \eta_{\text{init}} + \eta_{\text{cal}}. \tag{11}$$

The last terms are actual initialization and calibrated law-error allowances, not presumed zero. Timeout is heralded separately. The ideal hidden marginal is the passive waiting/window marginal evolved from the actual pretrial state; it is not silently changed to the backacted meter-end marginal.

With $b \asymp \ell$, the Brownian term in (6) suggests $b = O(\epsilon_m^2/\gamma)$ for a momentum-mark error ϵ_m, subject also to $B_* b/\sigma = O(\epsilon_m)$. A sufficient position accuracy is $e_x = O(\sigma b \epsilon_m/M)$. For white position noise a sufficient density is $\nu_x = O(\sigma^2 b^3 \epsilon_m^2/M^2)$, hence the severe $O(\epsilon_m^8)$ dependence after the Brownian time choice. The pointer implementation replaces an arbitrary independent noise budget by the actual tradeoff (8)–(9). These are sufficient scalings, not minimax metrology laws.

Per trial the source remains occupied for at most $H + t_*+$ finite latency, future experiment duration and any nonoverlapped reset. The clock waiting is $O(D_0 \log(1/\alpha_c)/(\gamma\delta_{\text{th}}^2))$. Temporary receiver memory can be logarithmic in ADC range, target rounding and clock range. Clipped marking

uses $O(B^2)$ elementary bit work at B-bit precision with ordinary arithmetic, plus the actual sensing/clock/output processing. A ticking watchdog pays every tick; asynchronous timing does not erase standby power.

Receiver memory has its own finite initialization, retention and reset contracts. For example, thermally activated bits with idle rate bounded by $\nu_m e^{-E_b/(k_B T_m)}$ can keep b_m bits over a finite duration H_m at failure δ_m with $E_b \geq k_B T_m \max\{0, \log(b_m \nu_m H_m/\delta_m)\}$. A known isolated two-state reset with energy bias Δ and relaxation rate κ_m uses $\Delta \geq k_B T_m \log(2b_m/\delta_r)$ and $t_r \geq \kappa_m^{-1} \log(2b_m/\delta_r)$. Raising the unwanted-state energy costs at most $b_m \Delta$ positive work in that model; switching and driver costs add. This is a finite-device upper-cost contract, not an assertion that Landauer's lower bound is attained in finite time.

7. Evaluation and reuse

If the future channel ignores the arm, the causal contrast for a bounded outcome is exactly

$$\tau = \mathbb{E}_{q_+} Y - \mathbb{E}_{q_-} Y = 2\mathbb{E}[s(p)Y]. \tag{12}$$

This ordinary weighted estimator removes the coin and has variance no larger than $2AY$; the difference is $4\mathbb{E}[(1-s^2)Y^2]$. A finite private weight cache must be withheld from the common future controller. After outcome measurement, retain a quantized score and erase or isolate the cache. The companion proves an explicit bias budget with $O(\log(1/\epsilon))$ weight and score bits. If the arm selects a later physical action, use the actual arm instrument instead.

For reuse, let R_j be all actually retained records. Appending a b-bit report adds $I(S : Y \mid R_j) \leq b \log 2$ to the joint entropy budget. Momentum-force action contributes a conservative $\mathbb{E}\int u^2 dt/(2\gamma\sigma^2)$; Girsanov improves the coefficient to $1/4$ at a fixed or source-independent endpoint. A state-dependent stopping time does not inherit the stationary-endpoint argument. Pad to a fixed deadline or use a separate proved bound. Work is not substituted for force-squared action for arbitrary feedback.

An actual-history Markov-kernel hybrid gives total campaign TV at most $\sum_j \epsilon_j$. It applies entropy only on actual histories, not an unproved ideal-history budget. With n rounds, total path error ϵ, and budgets D_j, comparable per-round errors give a waiting bill of order $n^2 \sum_j D_j \log(n/\alpha)/(\gamma\epsilon^2)$, which can be very large.

For a scalar causal estimate rather than a complete path-law comparison, set $X_j = 2A_j Y_j \in [-2, 2]$. Let τ_j be the ideal contrast for the current actual pretrial history and its passive-window hidden law. If the one-round TV errors have average bounds ϵ_j, Markov plus a bounded martingale inequality gives, with probability at least $1 - \alpha - \eta$,

$$\left| \frac{1}{n}\sum_j X_j - \frac{1}{n}\sum_j \tau_j \right| \leq \sqrt{\frac{8\log(2/\alpha)}{n}} + \frac{4}{\eta}\frac{1}{n}\sum_j \epsilon_j. \tag{13}$$

This is a current-environment causal contrast, not an iid or stationary population estimate. Average joint entropy is not a uniform bound on each rare history. Retained finite reports, imperfect reset residues, actuation and future outcome exposure all remain paid.

8. A concrete separation from global mixing

Consider two coupled rotors with

$$U(q,y) = B\sin^2 y + g[1-\cos(q-y)],$$

momenta p, r, masses M, m, and a thermostat only on p. The local force has magnitude at most g, independently of B. Let the initial source be Gibbs conditioned on the left hidden well, and shift p by $a\sigma R$ with a fair retained sign R. Then

$$D_0 = \log 2 + a^2/2, \qquad \left\| \frac{d\rho_0}{d(\pi\rho_R)} \right\|_2^2 = 2e^{a^2}.$$

Writing $b = \beta B$ and $Z_y = 2\pi e^{-b/2} I_0(b/2)$, the exact stationary outward crossing flux is

$$\lambda_B = \frac{2e^{-b}}{Z_y\sqrt{2\pi\beta m}}.$$

A crossing-count bound and Cauchy–Schwarz change of initial measure imply

$$\mathrm{TV}(\rho_t, \pi) \geq \frac{1}{2} - \sqrt{2e^{a^2}\lambda_B t}.$$

Thus global TV is at least $1/4$ through

$$t_B = \frac{Z_y\sqrt{2\pi\beta m}\, e^{\beta B}}{64e^{a^2}} = \Theta\left(\frac{e^{\beta B}}{\sqrt{B}}\right). \tag{14}$$

The local clock bound has no B dependence. At unit β, m, γ, a, the clean uniform-clock one-percent local bound is 2982.87, while $\beta B = 30$ gives $t_B = 1.00518 \times 10^{11}$. These are formula evaluations before meter and timing implementation allowances, not measured laboratory times.

Initialization is a real source resource. The exact conditioned ensemble is not supplied by waiting cheaply for global equilibrium. A broader narrow-well density family of width $O(B^{-1/2})$ has constant KL and requires finite-density loading control, $O(\log B)$ command bits and hidden factory access. Its factory cost is separate from deployment access. An angular encoder is also required for rotor readout. Equation (14) proves a separation of operations; it does not prove an exponential advantage over the best equally informed local method.

9. A boundary for continuous nonthermal driving

For ongoing forcing u in the noisy momentum direction,

$$\frac{1}{T}\int_0^T J(t)dt \leq \frac{D_0}{\gamma T} + \frac{\mathbb{E}\int_0^T u^2 dt}{2\gamma^2\sigma^2 T}. \tag{15}$$

This follows from entropy balance and Young's inequality. A driven free rotor with constant torque u has stationary momentum $N(u/\gamma, \sigma^2)$ and exactly $J = u^2/(2\gamma^2\sigma^2)$. Its distance from the desired Maxwell marginal is $2\Phi(|u|/(2\gamma\sigma)) - 1$ at every independent trigger time. A marking-only all-runs instrument leaves that marginal unchanged. Longer waiting cannot remove this floor. A real actuator cancelling the drive changes the interface. Continuous equilibrium heat and persistent nonthermal work are different resources.

10. Independent companions, execution and limits

The decision constructor proved an exact Blackwell ordering for known ordered LTI binary experiments and an unknown-initial-state annihilator attaining the $\lambda^{-2/3}$ compliance-decision scale. Numerical annihilation residual was 2.61×10^{-10} and the forcing identity relative error 6.60×10^{-13}. This is a completed structured decision, without a claim of advantage over equally informed matched testing.

The preparation-uncertainty constructor proved an exact conservative-completion Gram kernel and finite-record minimax radius. At the inherited acquired table precision, its sufficient short-box radius 4.8732 is worse than the energy radius 1; it correctly falls back. The theorem remains valid, but that numerical point is not a useful new predictor.

Gaussian-integral checks verify the arm means, equal kinetic energies and the weighted-score variance identity. Sinusoidal and clipped-linear arm information is respectively 0.26175224 and 0.33389329 nats at unit scale. The rotor programme evaluates the exact flux comparison. These are analytic/floating checks, not a laboratory execution or proof by simulation. The separate finite receiver illustration now executes 100,000 independently initialized linear-source trials, including smooth pointer/source dynamics, 32-bit position ADCs, a clipped-linear probability, a 24-bit coin and a finite future outcome. It produced 99,070 clock successes, 930 timeouts and no ADC overflows. Its derived conditional joint error allocation is 0.0467644 before unexecuted hardware discrepancies: 0.03 thermal, 0.01 pointer backaction, 0.00676438 metrology, plus coin and range terms. The maximum physical clock wait per attempt is 1292.14 and the expected clock occupancy is 277.778 in the chosen units; meter duration is 0.0002. The source retained strong hidden/reference correlation (sample normalized score 0.8985), while the arm/hidden score was 0.00274 with Monte Carlo standard error 0.00428. Actual and ideal causal scores were 0.14125 and 0.14265, with standard errors about 0.00447 each. These checks do not certify full total variation.

The simulation samples long Gaussian transitions numerically; it does not physically skip waiting. Serial clock occupancy for its 100,000 attempts would have expectation 27,777,778 time units, before initialization and outcome work. Cold pointers, source factory, clock privacy, finite release/reset, calibration, physical entropy bits and quantum/classical validity were not experimentally executed. The main nonlinear theorem stands independently of this linear illustration and does not acquire its premises from those samples.

The earlier complete predictive-preparation archive is preserved byte-for-byte, including nested earlier results. This thermal theorem does not generalize the finite-total-energy loaded LTI theorem by deleting its assumptions. The nonlinear positive-relaxation bridge and the acquired thermal feedback simulator remain separate operations. The earlier quantum results retain their purifiers, environment access, memory and half-diamond conventions. No unresolved weak partial-SWAP receiver is resolved here.

11. Sources and strongest comparisons

The closest entropy machinery is Vladimirov's conditional momentum decomposition [1], Desvillettes–Villani's velocity-dissipation analysis [2], and Gaussian logarithmic Sobolev/hypocoercivity. Recent global kinetic results [3,4] use additional spatial assumptions. The contribution claimed here is the finite clock/reference/meter/instrument construction built from that machinery.

Andersen reset [5] is faster if actual independent momentum overwrite exists. Isoconfigurational simulation [6] has stronger microscopic preparation access. Ordinary random force trials are already causal but target a different input law. Weighted change of measure is incorporated whenever it is the stronger estimator. Measurement-noise constraints [7] and the finite classical thermal range [8] remain substantive. No rare-event or universal algorithmic speedup is inferred from these comparisons.

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   (14) uses its own direct flux proof, not an imported asymptotic rate law.
10. Governing Astra predictive-preparation package, pinned revision 552a3c6: complete public governing source. Full revision identity is retained in the source receipt.
11. Accepted acquired thermal simulator N1688, pinned revision 626bf0f: complete accepted source.

The reproducible archive contains full subsidiary proofs, execution receipts, source comparisons, preservation hashes and the unchanged earlier package.